

Git & GitHub Cheat Sheet

The words

Repository	A project folder that remembers every version
Commit	A checkpoint, with a note saying why
Branch	A safe place to be wrong
main	The branch that should always work
Fork	Copy a repo into your own GitHub account
Clone	Copy a repo down onto your computer
Push	Send your commits up to GitHub
Pull	Bring everyone else's commits down to you
Pull request	Ask a maintainer to take your changes
Merge	Your changes become part of the project

One-time setup

```
git config --global user.name "Your Name"
git config --global user.email "you@example.com"
```

Use the email on your GitHub account.

The everyday loop

git pull	Get latest before starting
git status	Where am I? What changed?
git add .	Stage everything changed
git commit -m "..."	Checkpoint with a reason
git push	Send it up to GitHub

Branching

git branch	List branches (* = now)
git checkout -b my-fix	New branch, switch to it
git checkout main	Switch back to main
git log --oneline	History, newest first

Your first contribution (browser)

- 1 Fork the repo — button, top right
- 2 Create a branch named after yourself
- 3 Add or edit your file in the GitHub editor
- 4 Commit with a message that explains the change
- 5 Open a pull request against the original repo
- 6 Reply to review comments, wait for the merge

Same thing with git installed

```
git clone <fork-url>    After forking on GitHub
git checkout -b my-name  Make your branch
git add .                Stage your edit
git commit -m "Add me"   Checkpoint it
git push origin my-name  Send the branch up
```

Commit messages

Finish this sentence: "If applied, this commit will ..."

Bad	update · fix · changes · final v2
Good	Add Anjula to contributors page Fix broken link in README

When something goes wrong

git status	Always start here
git diff	Exactly what you changed
git checkout -- <file>	Undo changes to a file
git log --oneline	Find a commit to revisit

Habits worth keeping

- Pull before you start working. Every time.
- Commit small and often, not once at the end.
- Never commit passwords, keys or .env files.
- Use .gitignore for node_modules and build output.
- Ask on the issue before starting work on it.